

Vishal Nagamalla

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EDUCATION

Rutgers University, School of Arts and Sciences: Dean's List

New Brunswick, NJ

B.S. in Computer Science and Data Science - **GPA: 3.89**

December 2026

- *Activities/Societies:* Robotics Team (Captain), Competitive Programming Team, AI Club, Data Science Club, Aresty RAD research

WORK EXPERIENCE

Applied Materials

Santa Clara, CA

Software Engineering Intern

MAY 2026 – AUGUST 2026

- Shipped a production **semiconductor chip inspection** platform used by **leading manufacturers** to catch defects **before costly production**.
- Modernized review tooling with an interactive **3D chip model (React Three)**, replacing legacy data tables and cutting review time by **60%**.
- Designed a **multi-chip** comparison tool; highlights differences, supports **data tuning**, and cuts user error by **40%** with **zero server calls**.
- Reverse-engineered an undocumented **auth and locking system** from decompiled source, delivering the first write path into a locked service.
- Built a **Java / Spring Boot** backend aggregating multi-service **gRPC** data, swappable between **various data sources with zero code changes**.
- Deployed **Docker images** to a **Kubernetes cluster** with isolated instances, automated rollouts, config management, and **instant rollback**.
- Engineered a **config-driven tier system** gating features and data access per customer, opening a paid tier projected to lift account value **~15%**.
- Integrated an **agentic voice AI assistant (LLM + speech-to-text)** letting engineers query and operate the app **hands-free** in plain language.
- Deployed a **self-hosted, on-prem LLM (vLLM)** with a **custom tool-calling layer** grounded in live app state over **secure JWT** streaming.

PRACSYS Lab: Physical AI

New Brunswick, NJ

Software Engineering Intern

APRIL 2025 – SEPTEMBER 2025

- Designed AI robotic motion systems in **Python/ROS**, implementing frameworks under the **GELLO imitation-learning** architecture.
- Deployed **supervised learning pipelines** for a 6-DOF robotic arm, streamlining data trajectory for **training efficiency by 35%**.
- Engineered **reinforcement, diffusion, and flow-matching** models in **PyTorch** to optimize motion policies under noisy environments.
- Conducted joint-state noise experiments enabling adaptive correction logic that improved **path stability by 22%**.
- Constructed MuJoCo simulations and benchmarking tools for **testing, visualization, and real-time debugging** of robotic behaviors.

Atyeti Inc.

Princeton, NJ

Software Engineering Intern

MAY 2024 – AUGUST 2024

- Automated data pipelines (**Python**) cleaning and validating for financial clients, cutting manual reformatting by **18 hrs/proj** at **97% accuracy**.
- Deployed inference logic as **containerized microservices** using Docker for scalable and reliable production deployment.
- Collaborated to implement **CI/CD deployment automation**, ensuring reproducible builds and version-controlled releases.

PROJECTS

RealEstatePro – Full-Stack, Admin CMS, Auth, Maps, S3 Uploads

Next.js, React, Neon Postgres, AWS S3 | January 2026

Production Realtor Portfolio Website

<https://real-estate-pro-iota.vercel.app>

- Built **production-grade listings + detail UX** with type filtering, **Favorites**, responsive cards, and **Leaflet maps** powered by geocoded pins.
- Developed a secure **Admin CMS** using **NextAuth** with **allowlist** to manage end-to-end **CRUD** for listings, rich metadata, and **photo ordering**.
- Designed a **Postgres schema** and **Next.js API Routes** with **server-side validation**, clean errors, and transactional writes for admin ops.
- Implemented **address-to-coordinates geocoding (Photon)** to auto-generate map pins and re-geocode on edits, keeping search/maps in sync.
- Shipped **direct-to-S3 uploads** via **presigned URLs** with **trusted-origin/CORS hardening**, ordered galleries, and safe server-side key checks.
- Deployed on **Vercel** with **Neon Postgres**, **production environment config (DB/Auth/AWS)**, and **dynamic API routes** to avoid build failures.

AirNote – FullStack, AI Integration, Gemini API, Computer Vision

Python, React, Firebase, Gemini API | October 2025

- Built a **scalable AI note management platform** using **Gemini, Firebase, and React** for automated labels, summaries, and categorization.
- Integrated **AR glasses** with **OpenCV** and an **AI voice assistant** for **gesture-based** note-taking and **real-world** contextual interaction.
- Developed a deployable **FastAPI backend** enabling real-time **ingestion, classification, and data storage** via **Firebase Cloud**.
- Automated **AI embedding and clustering** workflows to link related notes via **semantic analysis** for efficient retrieval and visualization.
- Designed a responsive **React/Tailwind UI** featuring live previews, **AI summaries**, and integrated testing for **reliability and scalability**.

HoloBrain (MLH WINNER) – ML, Computer Vision, Unity 3D Hologram

PyTorch, Python, C# || October 2024

- Interactive **3D hologram** for MRI brain scan visualization, real-time gesture control, using **Unity 3D, Sockets, and Python-based AI**.
- Established a CNN model with **97% accuracy** to classify healthy vs. unhealthy MRI scans, expediting the identification of critical cases.
- Dynamic **hand gesture tracking** using **OpenCV** and **MediaPipe** to manipulate holograms, enhancing interactivity and usability.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, SQL, HTML5, CSS3, R, $\LaTeX$, Bash/Shell, JSON/YAML

Frameworks/Libraries: Next/React/Node.js, FastAPI, Flask, Firebase, ASP.NET Core, Spring Boot, Tailwind CSS, React Three Fiber, Leaflet

Technologies: PostgreSQL, AWS S3, NextAuth, gRPC, PyTorch, OpenCV, MediaPipe, Gemini/OpenAI API, scikit-learn, Docker, Kubernetes, CI/CD